PV-RS-01 RS

Portview

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Revision History

Rev#	Description	Date	Remark
0.0	Initial version created after requirement subdivision	2023-12-11	
0.1	Added additional requirement specifications including translation support	2025-07-24	

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Document Overview

This document defines the system-level requirement set for Portview.

Document notes:

- The current draft is focused on the Portview viewer and supporting workflow functions.
- ImageProcess-specific requirement content is intentionally excluded from this document set.

1. Purpose

This document defines the system-level requirements for the Portview dental image viewer.

The requirement set is intended to:

- define the intended user-visible capabilities of the workstation software
- identify the main operating constraints and support functions
- provide the system-level input used by downstream software requirements, design, and verification documents

2. Scope

This document applies to the Portview software environment running on a Windows-based workstation.

In scope:

- image acquisition from supported sources
- patient and account handling
- image viewing, comparison, annotation, and measurement
- export, print, and DICOM communication
- device status, logging, and multilingual user interface behavior

Out of scope:

- ImageProcess-specific image-quality processing requirements

3. Requirement Summary

The system-level requirements currently fall into the following functional groups.

Functional Group	Coverage
Operating context	Windows account execution, workstation behavior, status messaging
Image acquisition	Intraoral sensor image capture, digital image capture, TWAIN import
Viewer functions	Main viewer, compare viewer, tooth map, pan, rotate, zoom
Output and communication	Print, export to file or media, DICOM transmission
Device interaction	Sensor selection, connection status, device status, FMX continuous acquisition
Data integrity and robustness	USB integrity confirmation, network support, activity logging, stability guidance
Administrative functions	Account management, annotation and measurement, multilingual GUI, error messaging

4. Requirement Catalogue

4.1 Operating Environment And Access

Requirement ID	Requirement	Acceptance Criteria
`RS-001`	Portview runs under a Windows OS account	Application launches and operates under a standard Windows user account without requiring administrative privileges for normal workflow
`RS-002`	Portview uses pop-up windows to present message or status information	Status and error messages are displayed as user-visible pop-up windows; messages remain visible until user acknowledgment

4.2 Acquisition And Device Interaction

Requirement ID	Requirement	Acceptance Criteria
`RS-003`	Acquire intraoral sensor image	Sensor image is acquired and displayed under the selected patient context without data corruption
`RS-004`	Acquire digital images	Digital image file is imported and associated with the active patient record
`RS-005`	Acquire image from TWAIN device	TWAIN image is acquired and displayed correctly in the viewer
`RS-006`	Provide FMX continuous acquisition	Sequential acquisition proceeds through all positions in the selected FMX layout
`RS-007`	Select one sensor from multiple available sensors	Selected sensor becomes active; subsequent acquisition uses the chosen device
`RS-008`	Display sensor connection status	Connection status changes between connected and disconnected states are reflected in the UI within the current workflow cycle
`RS-009`	Show device status	Current device state is displayed accurately for connected modalities

4.3 Viewing And Diagnostic Interaction

Requirement ID	Requirement	Acceptance Criteria
`RS-010`	Main viewer displays images on the mounted layout	Selected patient images are rendered on the correct mount layout without misattribution
`RS-011`	Provide tooth map window	Tooth map navigation selects and displays the correct image for the chosen position

Requirement ID	Requirement	Acceptance Criteria
`RS-012`	Provide image comparison	At least two images can be compared side-by-side with independent viewing state
`RS-013`	Allow image move, rotate, or zoom	Pan, rotate, and zoom operations update the display without losing image context or annotation state
`RS-014`	Provide annotation and measurement functions	Annotation and measurement tools create, display, and persist overlays correctly across viewer sessions

4.4 Output And Communication

Requirement ID	Requirement	Acceptance Criteria
`RS-015`	Provide printing	Selected image is printed through the standard print path without application error
`RS-016`	Export images to file	Image export file is created at the target location with preserved attribution
`RS-017`	Export images to media device	Export to CD or equivalent media completes successfully or reports a user-visible error
`RS-018`	Send image to DICOM server	DICOM transmission completes successfully or failure is detected and reported to the user

4.5 Integrity, Robustness, And User Guidance

Requirement ID	Requirement	Acceptance Criteria
`RS-019`	Confirm USB integrity before delivery or installation use	Integrity check detects corruption on delivered USB media and reports the result

Requirement ID	Requirement	Acceptance Criteria
`RS-020`	Support network connection needed to receive image from the device	Network-based acquisition path remains functional when the local network connection is available
`RS-021`	Record program activity in a log	Auditable operations are recorded in the activity log with sufficient detail for troubleshooting
`RS-022`	Provide system-instability guidance	Error type is classified and troubleshooting instructions are presented to the user
`RS-023`	Classify and display malfunction-related error messages	Malfunction-related errors produce a user-visible classified error message

4.6 Administrative And Localization Support

Requirement ID	Requirement	Acceptance Criteria
`RS-024`	Manage account	Patient records can be created, modified, searched, and deleted with data consistency preserved
`RS-025`	Support multilingual GUI behavior	GUI text changes to the selected language without layout corruption or content loss

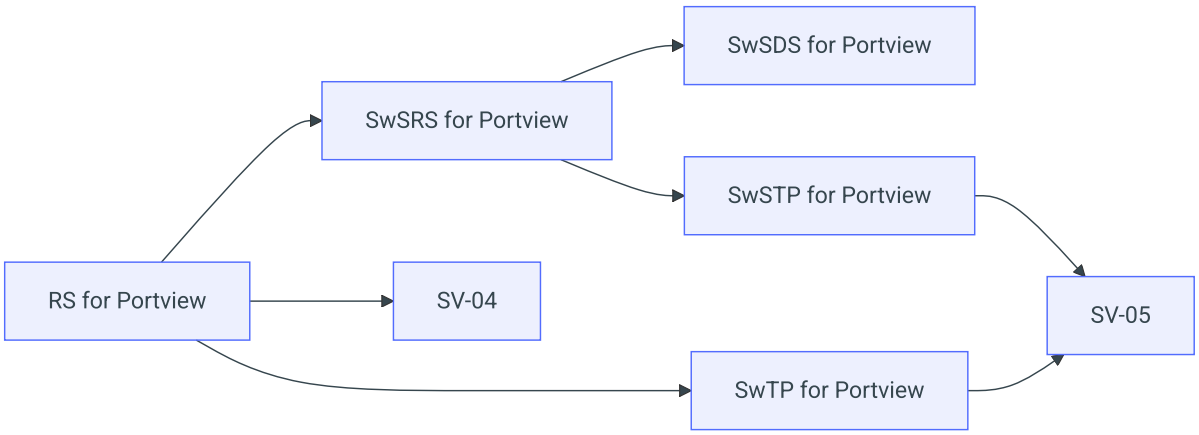
5. Interface Expectations

The system-level requirement set assumes the following external or user-facing interfaces:

- supported sensor or device interfaces used for image acquisition
- file-based and TWAIN-based import paths
- DICOM server or printer communication paths
- removable media or file-export destinations
- local workstation user-account and storage environment

6. Requirement Relationships

The system-level requirement set feeds the software requirement and verification set as shown below.



7. Verification Use

This requirement document is intended to feed:

- `SwSRS for Portview`
- `SwTP for Portview`
- `SV-04` Software Verification Plan
- `SV-05` Software Verification Report
- the Portview traceability matrix